MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

RECEIVER, INTERFACE, DIGITAL, KEY CODE JDK

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic interface receiver module.

1.2 Module description. The following electrical parameters reflect the -55° to 100°C end-of-life limits from table II.

Name	Receiver, interface
Family	Digital
Key code	JDK
Size	1A
Weight	35 grams maximum
Failure rate	
Class I	0.278 failures/10^6 hours maximum
Class II	2.395 failures/10^6 hours maximum
Environment	Classes I and II as specified in MIL-STD-1389
Supply voltage	5.0 Vdc ±10% 15.0 Vdc ±10% -5.0 Vdc ±5%
Maximum power dissipation	2.23 W (all outputs in low level) 2.0 W (all outputs in high level)
Input voltage	
Low level	1.8 V maximum
High level	3.2 V minimum
Input current	
Low level (V_{IL} = -2.0 V)	4.7 mA maximum
High level (V_{IH} = 15.0 V)	50 µA maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, DC 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

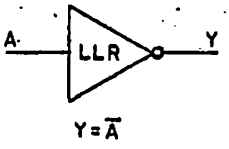
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Output voltage	
Low level	0.4 V maximum
High level	2.4 V minimum, V_{CC} maximum
Output current	
Low level (V_{OL} = 0.4 V)	12.8 mA maximum
High level (V_{OH} = 2.4 V)	320 µA minimum
Propagation delay time	
t_{PHL}	10 ns minimum, 37 ns maximum
t_{PLH}	10 ns minimum, 32 ns maximum

1.3 Module function. This module contains 18 receiver circuits as illustrated on figure 1. It is intended for use as a receiver when transmitting logic level signals from chassis to chassis within an equipment cabinet. When flat cable is used for interconnection, the length shall be less than 7 feet, and the driver and receiver shall be located adjacent to the ends of the cable. A dedicated driver shall be used to drive the interface receiver. A dedicated driver is defined as a circuit that drives no other circuits, either on its own module, or on another module except the interface receiver module. The driver circuit must be capable of driving at low level a -16 mA minimum at 0.4 V and at high level a +1.0 mA minimum at 2.4 V.

1.3.1 Input and output relationship. Each interface receiver shall have the following input and output relationship for positive logic:

TABLE I. Input and output for positive logic.



1.3.2 Parallel outputs. Paralleling of outputs is not permissible.

1.3.3 Crosstalk relationships. The module has the following crosstalk relationships.

1.3.3.1 Module crosstalk. When connected in the worst case mode of operation (all outputs but one switching simultaneously), the JDK module could exhibit substantial crosstalk. When a quiet output is a low level, there may be as much as an 820 mV positive going spike.

1.3.3.2 Circuit crosstalk. The crosstalk on any output pin is nearly directly proportional to the number of switching outputs. For every switching output in group A (circuits 1-4), the noise induced on a low level quiet output in group A is as much as 110 mV. For every switching output outside group A, the noise induced on a low level quiet output in group A is as much as 35 mV. The preceding relationship applies to group B (circuits 5-8), group C (circuits 9-11), group D (circuits 12-15), and group E (circuits 16-18).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

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STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.278 failures per million operating hours for class I and 2.395 for class II when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II and shall operate within the environment of design classes I and II as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be JDK.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/16-1 for class I and M28787/16-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table II.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table II(see 6.2).

3.9 Power supply voltage. The module shall conform to the requirements of table II for power supply voltages of +5.0 ±10 percent (V_{CC}), +15.0 V ±10 percent (V_{BB}), and -5.0 V ±5 percent (V_{EE}) when driving the specified loads in either logic state, except that the limits for output short circuit current apply only for a V_{CC} (+5 V) power supply voltage of +5.5 Vdc ±1 percent.

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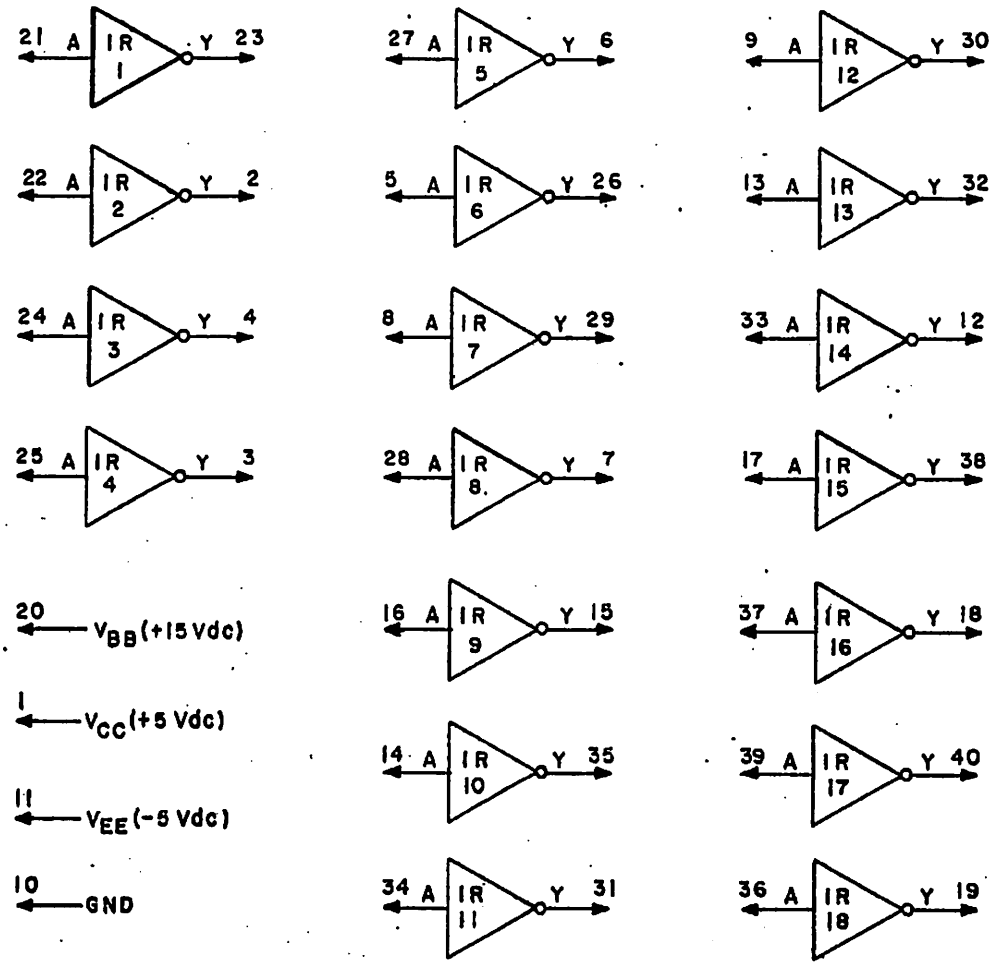


FIGURE 1. JDK module function diagram.

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TABLE II. Module electrical requirements.

Test cond	Parameter	Ref test table	Test type (see notes)	-55° ±5°C Initial & end-of-life limits		0°±5°C Initial & end-of-life limits		25° ±5°C Initial & end-of-life limits		60°±5°C Initial & end-of-life limits		100°±5°C Initial & end-of-life limits		Unit
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A18	V_{OUT} (low-loaded)	III	S	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	V
B	Isolation													
	Initial	IV	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	IV	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
C1-C18	Input capacitance	V	Q	2	20	2	20	2	20	2	20	2	20	pF
D1-D18	Static output voltage, low level	VI	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
E1-E18	Static output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F1-F18	Open circuit input voltage	VIII	X	4.1	5.3	4.1	5.3	4.1	5.3	4.1	5.3	4.1	5.3	V
G1-G18	Input current, low level	IX	X	2.7	4.7	2.7	4.7	2.7	4.7	2.7	4.7	2.7	4.7	mA
H1-H18	Input current, high level	X	X	0	50	0	50	0	50	0	50	0	50	µA
J1-J18	Output short circuit current	XI	X	23	55	23	55	23	55	23	55	23	55	mA
K1	Power supply current (V_{CC}), low level	XII	S	30	150	30	150	30	150	30	150	30	150	mA
K2	Power supply current (V_{CC}), high level	XII	S	5	30	5	30	5	30	5	30	5	30	mA
K3	Power supply current (V_{BB}), low level	XII	S	10	58	10	58	10	58	10	58	10	58	mA
K4	Power supply current (V_{BB}), high level	XII	S	15	84	15	84	15	84	15	84	15	84	mA
K5	Power supply current (V_{EE}), low level	XII	S	15	84	15	84	15	84	15	84	15	84	mA

See notes at end of table.

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TABLE II. Module electrical requirements -Continued.

Test cond	Parameter	Ref test table	Test type (see notes)	-55° +5°C Initial & end-of-life limits		0°+5°C Initial & end-of-life limits		25° +5°C Initial & end-of-life limits		60°+5°C Initial & end-of-life limits		100°+5°C Initial & end-of-life limits		Unit
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
K6	Power supply current (V_{EE}), high level	XII	S	15	84	15	84	15	84	15	84	15	84	mA
L1-L18	Transition time (t_{TLH})	XIII	S	20	78	20	78	20	72	20	78	20	78	ns
M1-M18	Transition time (t_{THL})	XIII	S	8	35	8	35	8	30	8	35	8	35	ns
N1-N18	Propagation delay time (t_{PHL})	XIV	S	10	37	10	37	10	35	10	37	10	37	ns
P1-P18	Propagation delay time (t_{PLH})	XIV	S	10	32	10	32	10	30	10	32	10	32	ns
R1-R18	Output crosstalk, low level	XV	Q	0	820	0	820	0	820	0	820	0	820	mV
S1-S18	Output crosstalk, high level	XVI	Q	0	400	0	400	0	400	0	400	0	400	mV
T1	Power supply transient amplitude (V_{CC})	XVII	Q	0	1.0	0	1.0	0	1.0	0	1.0	0	1.0	V
T2	Power supply transient amplitude (V_{BB})	XVII	Q	0	400	0	400	0	400	0	400	0	400	mV
T3	Power supply transient amplitude (V_{EE})	XVII	Q	0	400	0	400	0	400	0	400	0	400	mV
U1	Power supply transient rise time (V_{CC})	XVIII	S	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0	ms
U2	Power supply transient rise time (V_{BB})	XVIII	S	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0	ms
U3	Power supply transient rise time (V_{EE})	XVIII	S	← See figure 19 →										

See notes at end of table.

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TABLE II. Module electrical requirements -Continued.

NOTES:

1. All (X) parameters shall be 100 percent tested during module acceptance tests.
2. All (S) parameters shall be 25°C ambient sample acceptance tested (1 percent, or two modules from each inspection lot, whichever is greater).
3. All (Q) parameters shall be tested during qualification testing only.
4. Unless otherwise specified, limits are for initial and end-of-life.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
DC power supply	Item P01	5
DC voltmeter	Item V03	1
Megohm bridge	Item R06	1
Capacitance meter	Item C05	1
DC ammeter	Item X01	1
Pulse generator	Item J01	1
Oscilloscope	Item D05	1
Oscilloscope probe	Item Y01	2
Square wave generator	Item H01	1
Oscilloscope	Item D01	1
Oscilloscope probe	Item Y02	1
Oscilloscope probe	Item Y04	1
Resistors		
590Ω ±1%, 1/4 W		18
402Ω ±1%, 1/4 W		18
51Ω ±5%, 1/2 W		2
7.5 kΩ ±1%, 1/4 W		18
Value as required for figure 9, note 4, 1/4 W		18
511Ω ±1%, 1/4 W		1
Capacitors		
100 pF ±20%		18
Value as required for figure 9, note 1		18
Value as required for figure 9, note 8		18
Diode		
1N4150, or equivalent		18*

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table II.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program (SEMP) Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. If test conditions A and B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

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4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table II shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through P, and U, shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table II which are marked with an (X) in the test type column, and test conditions U1 through U3, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through P and U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table II. Following the return of the module to 25°C, electrical tests B through U shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table II.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life (see 3.8).

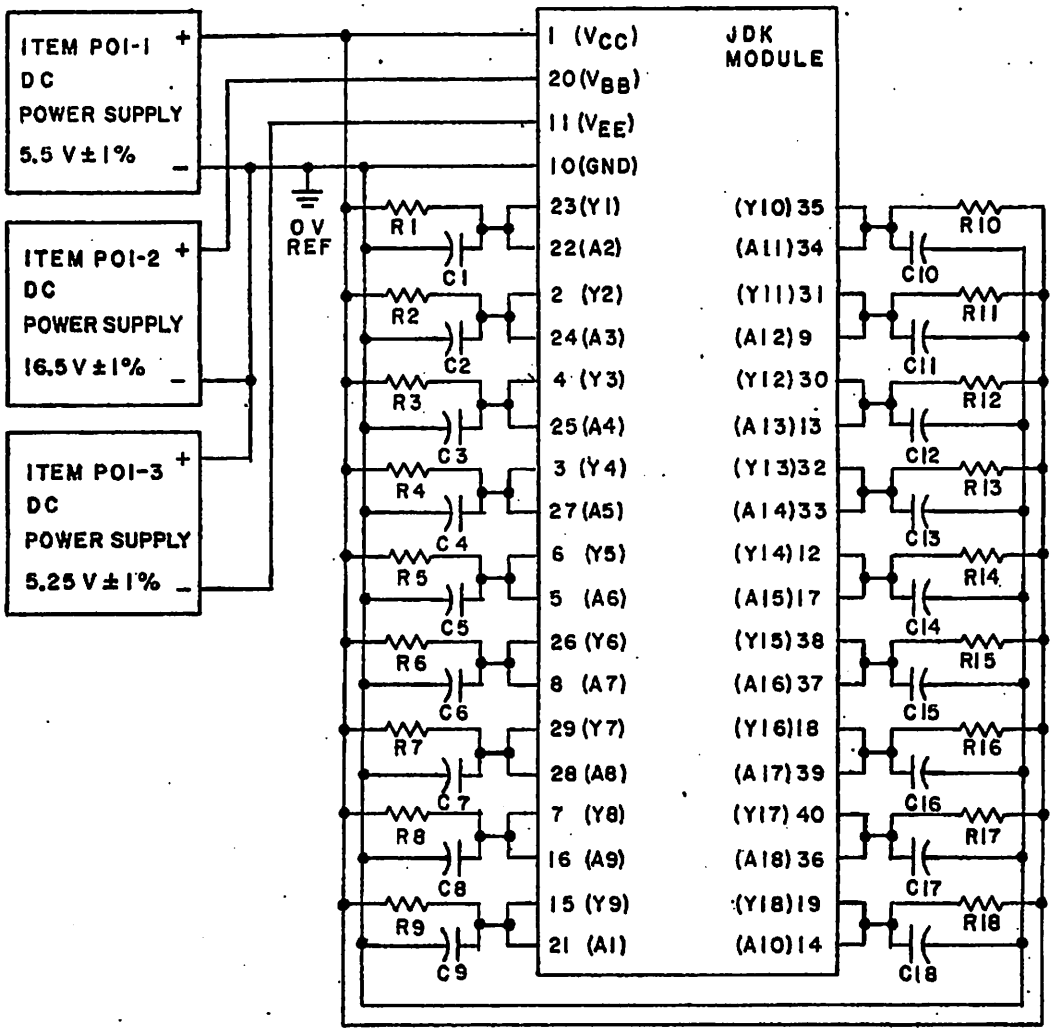
6.3 Cross-reference of classification. Part number M28787/16-1 supersedes part number M28787/16. The two parts are interchangeable.

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5963-N017)

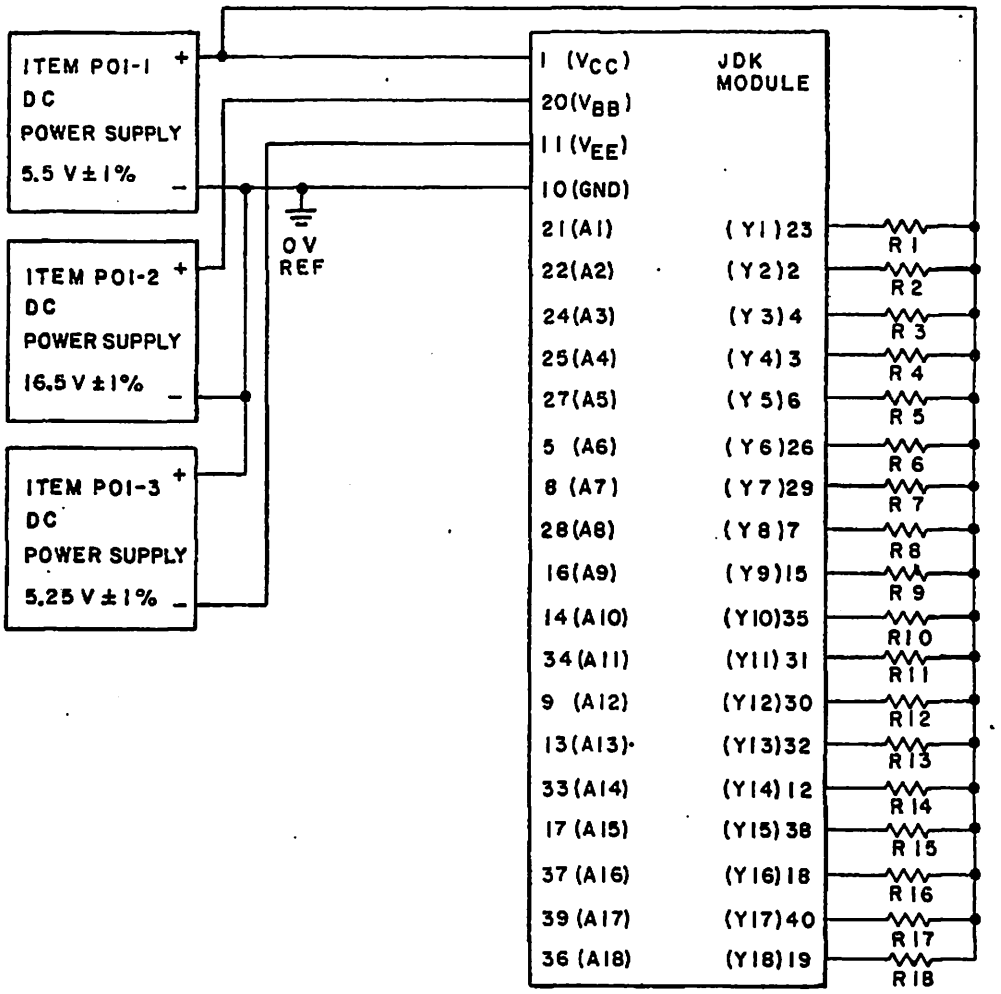
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NOTES:
1. R1-R18 = 590Ω ±1%, 1/4 W.
2. C1-C18 = 100 pF ±20%.
3. When using figure 2 for module burn-in, the RC loads are not required.

FIGURE 2. Operating temperature, burn-in and life test circuit.

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NOTE: R1-R18 = 402Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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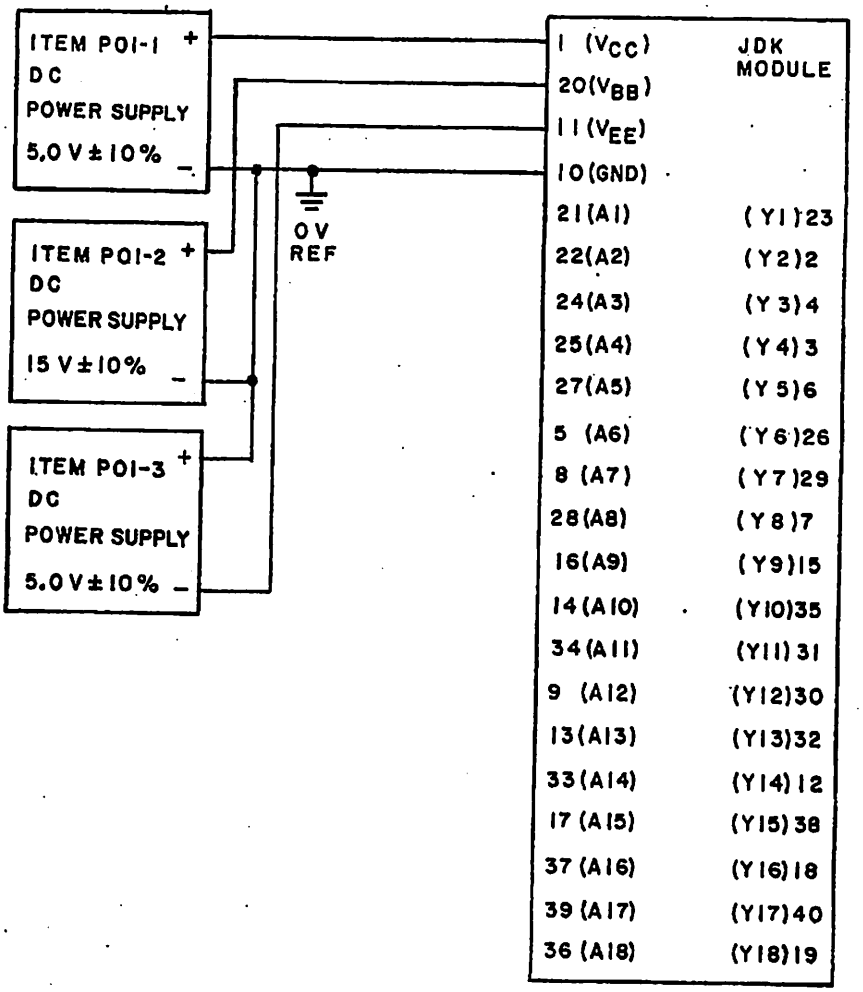


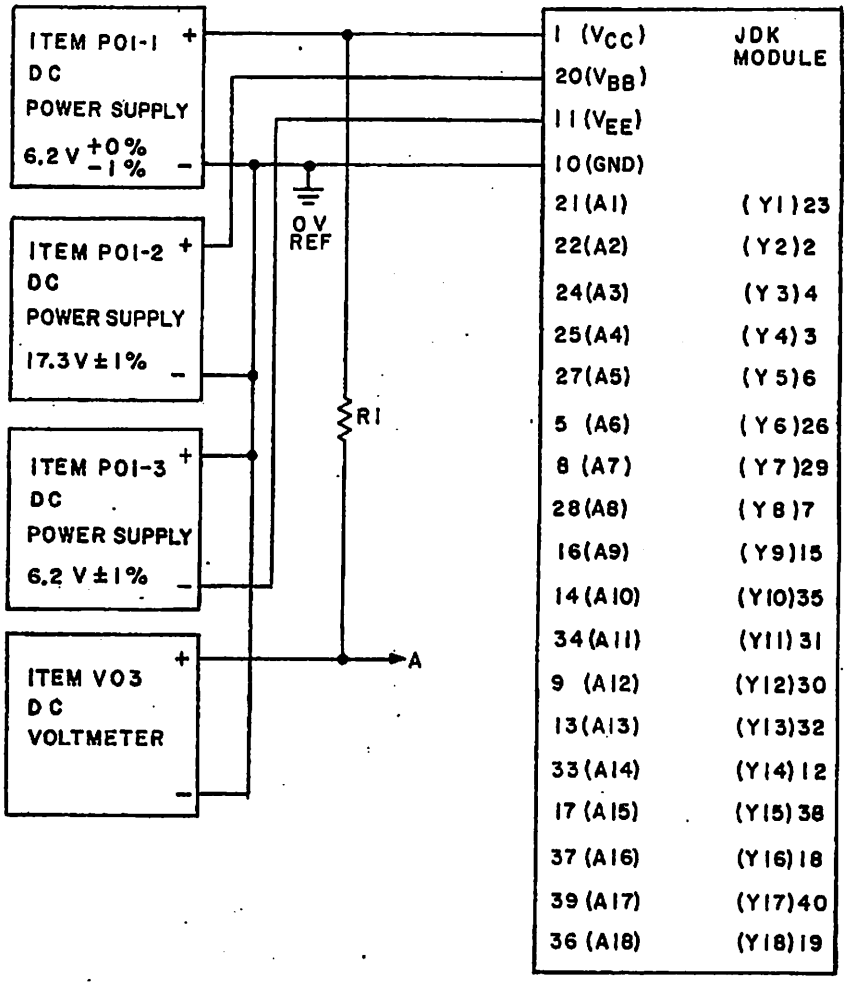
FIGURE 4. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
A1	Y1 V_{OUT} (low-loaded)	23	0.01	0.8	V	5
A2	Y2	2				
A3	Y3	4				
A4	Y4	3				
A5	Y5	6				
A6	Y6	26				
A7	Y7	29				
A8	Y8	7				
A9	Y9	15				
A10	Y10	35				
A11	Y11	31				
A12	Y12	30				
A13	Y13	32				
A14	Y14	12				
A15	Y15	38				
A16	Y16	18				
A17	Y17	40				
A18	Y18 V_{OUT} (low-loaded)	19	0.01	0.8	V	5

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NOTE: RI = 402Ω ±1%, 1/4 W.

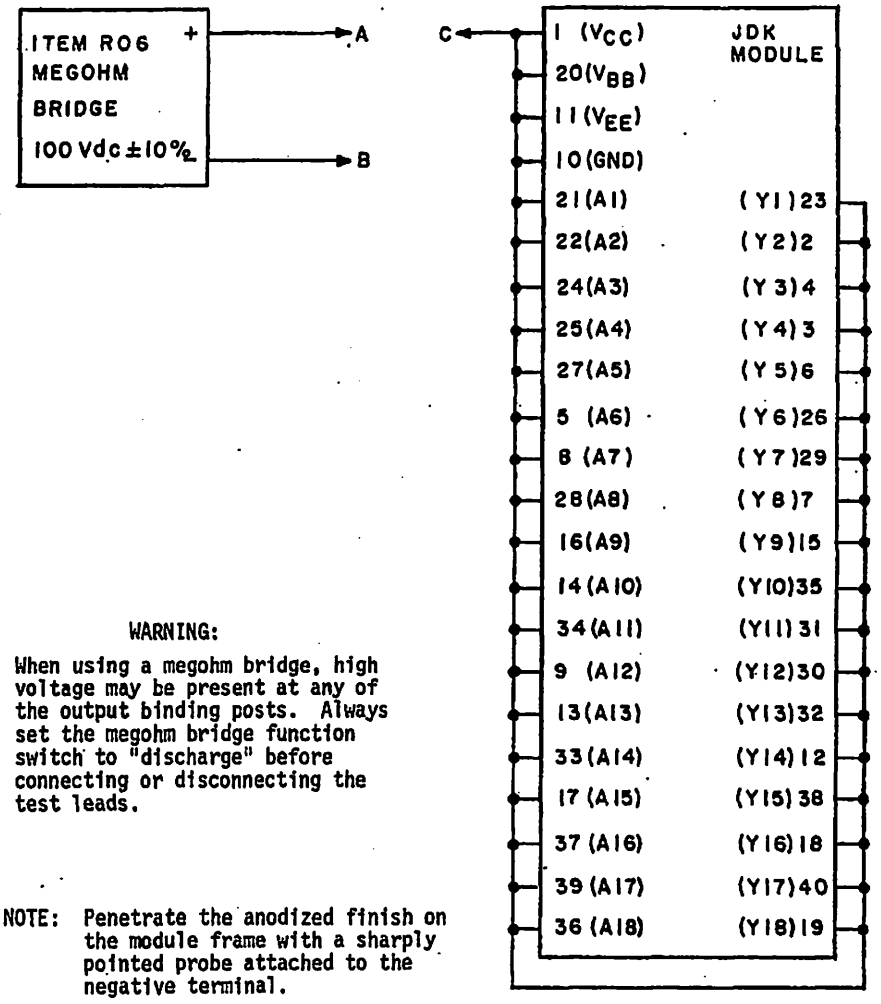
FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Isolation test.

Test condition	Parameter	Connect		Initial limits ?5° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B	Isolation	C	Frame	10	∞	MΩ	6

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe attached to the negative terminal.

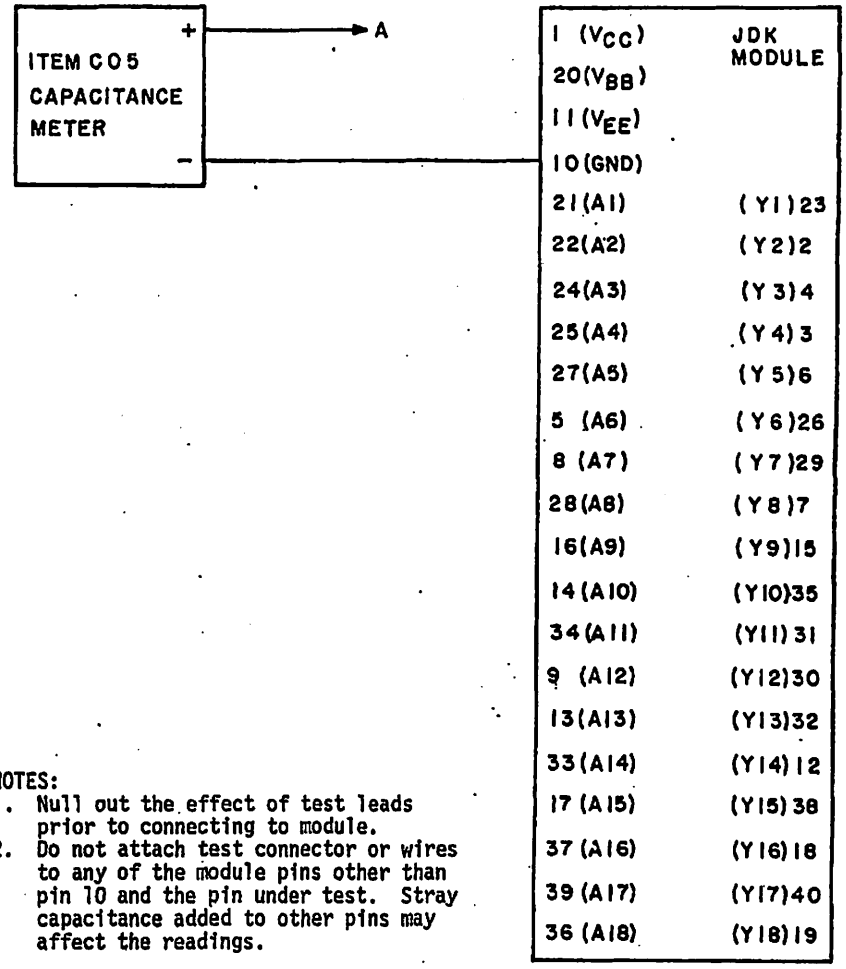
FIGURE 6. Isolation test circuit for table IV.

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TABLE V. Input capacitance test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
C1	A1 Input capacitance	21	2	20	pF	7
C2	A2	22				
C3	A3	24				
C4	A4	25				
C5	A5	27				
C6	A6	5				
C7	A7	8				
C8	A8	28				
C9	A9	16				
C10	A10	14				
C11	A11	34				
C12	A12	9				
C13	A13	13				
C14	A14	33				
C15	A15	17				
C16	A16	37				
C17	A17	39				
C18	A18 Input capacitance	36	2	20	pF	7

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NOTES:
1. Null out the effect of test leads prior to connecting to module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

FIGURE 7. Input capacitance test circuit for table V.

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TABLE VI. Static output voltage (low-level) test.

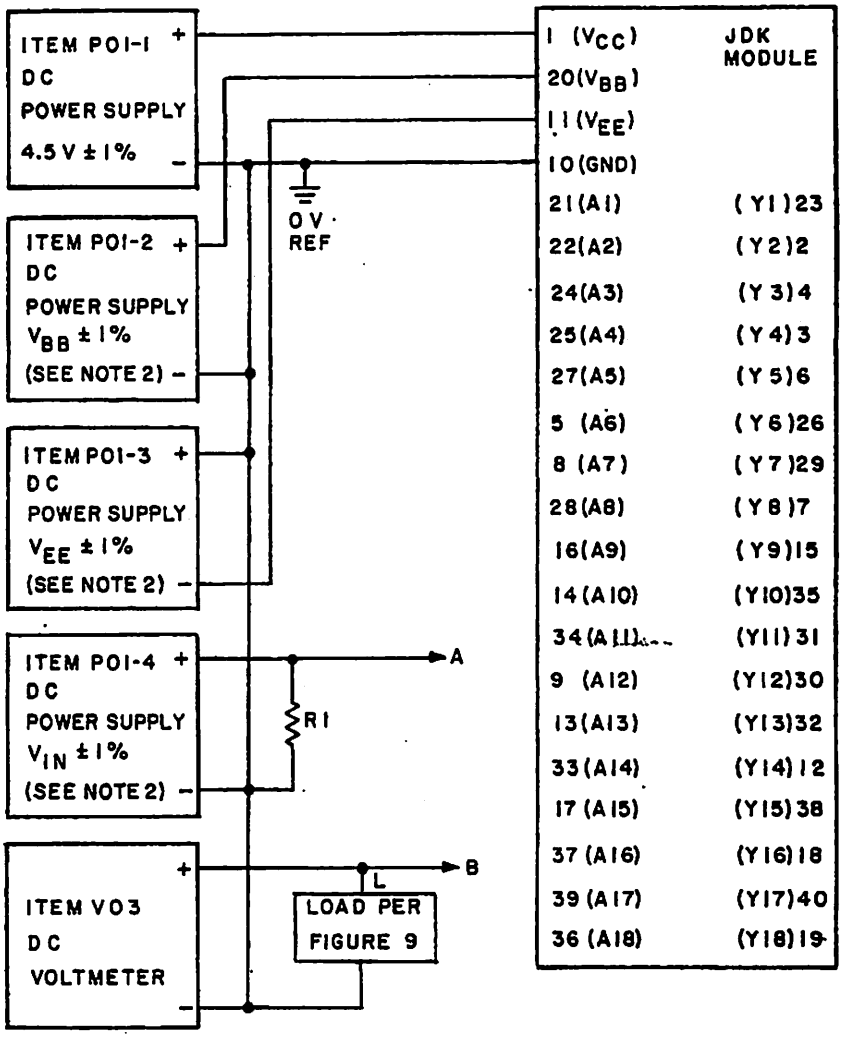
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Y1 Static output voltage, low level	21	23	0.0	0.4	V	8
D2	Y2	22	2				
D3	Y3	24	4				
D4	Y4	25	3				
D5	Y5	27	6				
D6	Y6	5	26				
D7	Y7	8	29				
D8	Y8	28	7				
D9	Y9	16	15				
D10	Y10	14	35				
D11	Y11	34	31				
D12	Y12	9	30				
D13	Y13	13	32				
D14	Y14	33	12				
D15	Y15	17	38				
D16	Y16	37	18				
D17	Y17	39	40				
D18	Y18 Static output voltage, low level	36	19	0.0	0.4	V	8

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TABLE VII. Static output voltage (high-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	Y1 Static output voltage, high level	21	23	2.4	4.5	V	8.
E2	Y2	22	2				
E3	Y3	24	4				
E4	Y4	25	3				
E5	Y5	27	6				
E6	Y6	5	26				
E7	Y7	8	29				
E8	Y8	28	7				
E9	Y9	16	15				
E10	Y10	14	35				
E11	Y11	34	31				
E12	Y12	9	30				
E13	Y13	13	32				
E14	Y14	33	12				
E15	Y15	17	38				
E16	Y16	37	18				
E17	Y17	39	40				
E18	Y18 Static output voltage, high level	36	19	2.4	4.5	V	8

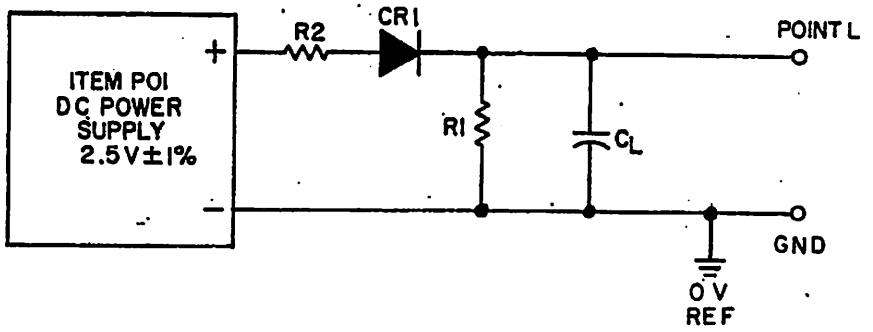
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. For test conditions D1-D18; V_{BB} = 13.5 V, V_{EE} = 5.25 V, V_{IN} = 3.2 V and for test conditions E1-E18; V_{BB} = 16.5 V, V_{EE} = 4.75 V, V_{IN} = 1.8 V.

FIGURE 8. Static output voltage (low-level and high-level) test circuit for tables VI and VII.

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NOTES:
1. C_L = 300 ±15 pF (including probe and parasitic capacitance) for test conditions L, M, N, and P.
2. CR1 = 1N4150, or equivalent.
3. R1 = 7.5 kΩ ±1%, 1/4 W.
4. R2 = Select value for current flow of 13.8 mA ±1% from point "L" to ground.
5. C_L is not required for dc electrical tests D1-D18 or E1-E18.
6. The maximum lead length from point "L" to the appropriate module pin shall be 3 inches (76.2 mm).
7. During output switching, power supply item P01 shall have a maximum overshoot of 500 mV.
8. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions R and S.

FIGURE 9. Logic load circuit.

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TABLE VIII. Open circuit input voltage test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
F1	A1 Open circuit input voltage	21	4.1	5.3	V	10
F2	A2	22				
F3	A3	24				
F4	A4	25				
F5	A5	27				
F6	A6	5				
F7	A7	8				
F8	A8	28				
F9	A9	16				
F10	A10	14				
F11	A11	34				
F12	A12	9				
F13	A13	13				
F14	A14	33				
F15	A15	17				
F16	A16	37				
F17	A17	39				
F18	A18 Open circuit input voltage	36	4.1	5.3	V	10

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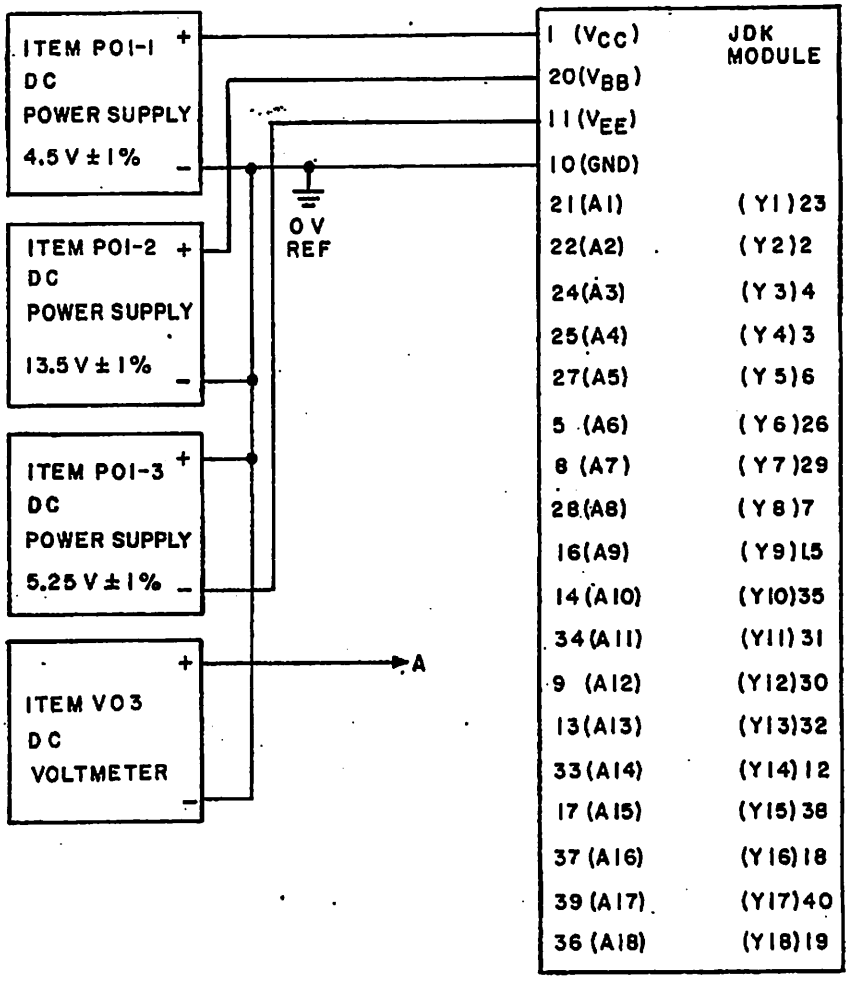


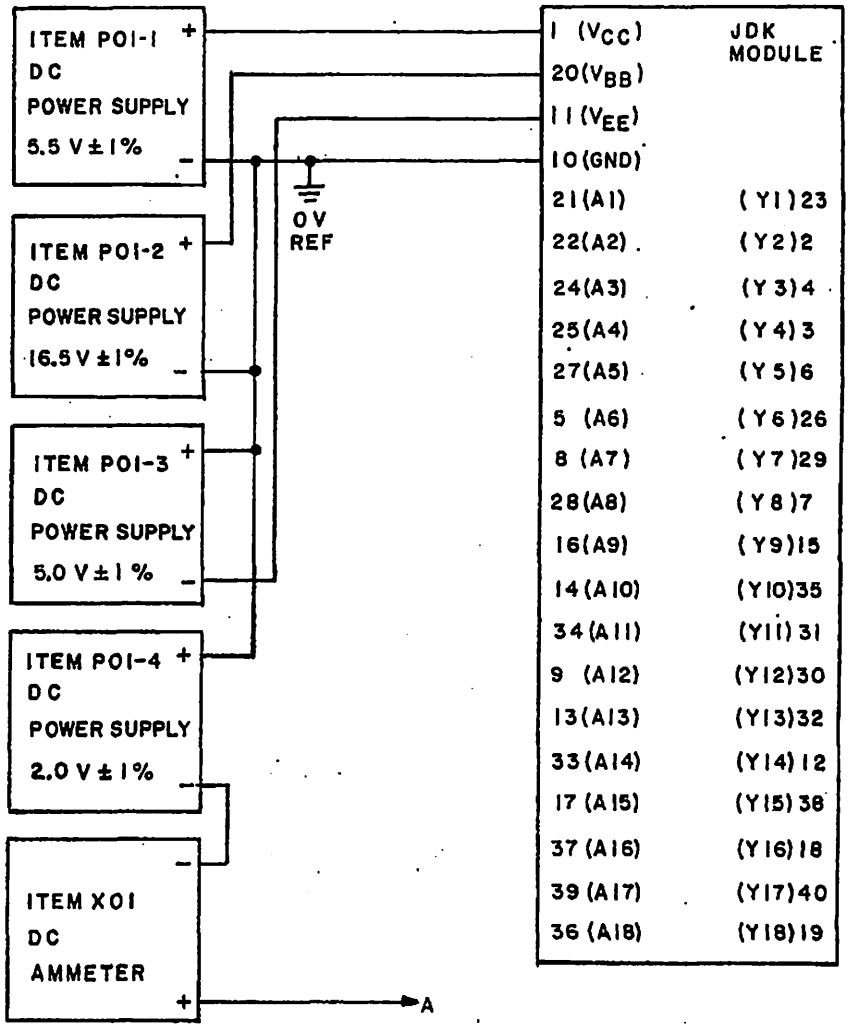
FIGURE 10. Open circuit input voltage test circuit for table VIII.

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TABLE IX. Input current (low-level) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
G1	A1 Input current, low level	21	2.7	4.7	mA	11
G2	A2	22				
G3	A3	24				
G4	A4	25				
G5	A5	27				
G6	A6	5				
G7	A7	8				
G8	A8	28				
G9	A9	16				
G10	A10	14				
G11	A11	34				
G12	A12	9				
G13	A13	13				
G14	A14	33				
G15	A15	17				
G16	A16	37				
G17	A17	39				
G18	A18 Input current, low level	36	2.7	4.7	mA	11

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CAUTION:
Do not isolate the chassis of ammeter X01 from earth ground.

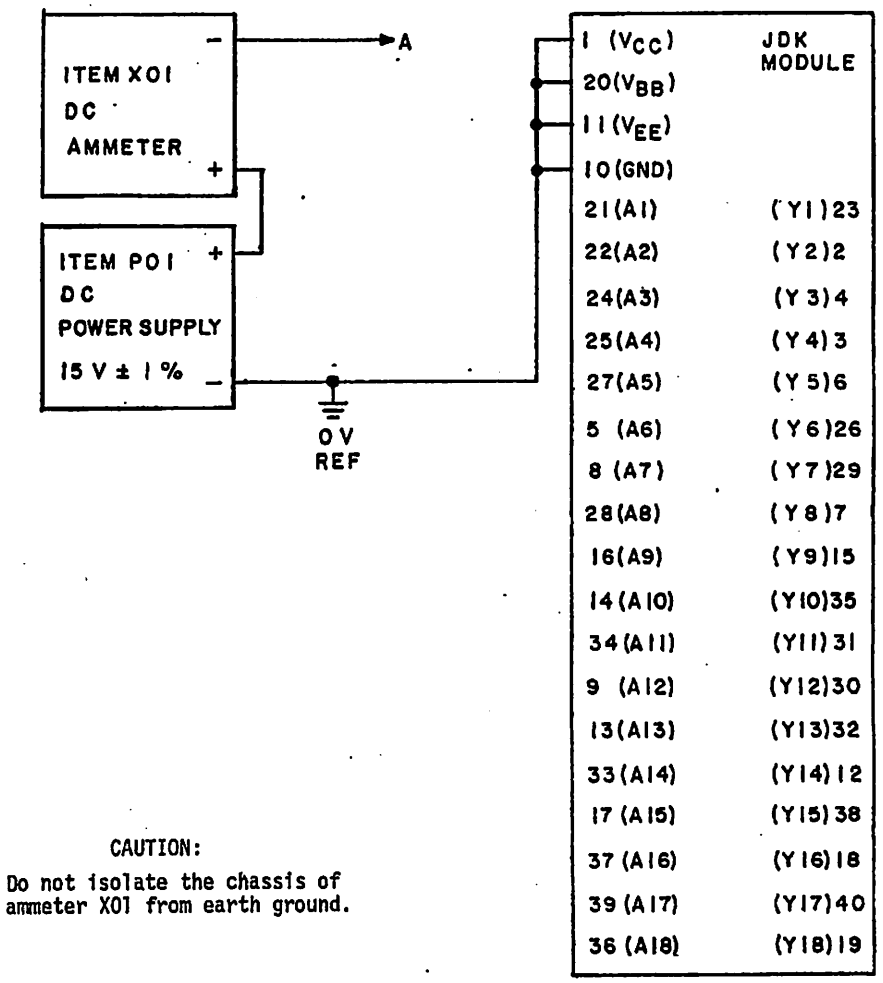
FIGURE 11. Input current (low-level) test circuit for table IX.

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TABLE X. Input current (high-level) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
H1	A1 Input current, high level	21	0	50	µA	12
H2	A2	22				
H3	A3	24				
H4	A4	25				
H5	A5	27				
H6	A6	5				
H7	A7	8				
H8	A8	28				
H9	A9	16				
H10	A10	14				
H11	A11	34				
H12	A12	9				
H13	A13	13				
H14	A14	33				
H15	A15	17				
H16	A16	37				
H17	A17	39				
H18	A18 Input current, high level	36	0	50	µA	12

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CAUTION:
Do not isolate the chassis of ammeter XO1 from earth ground.

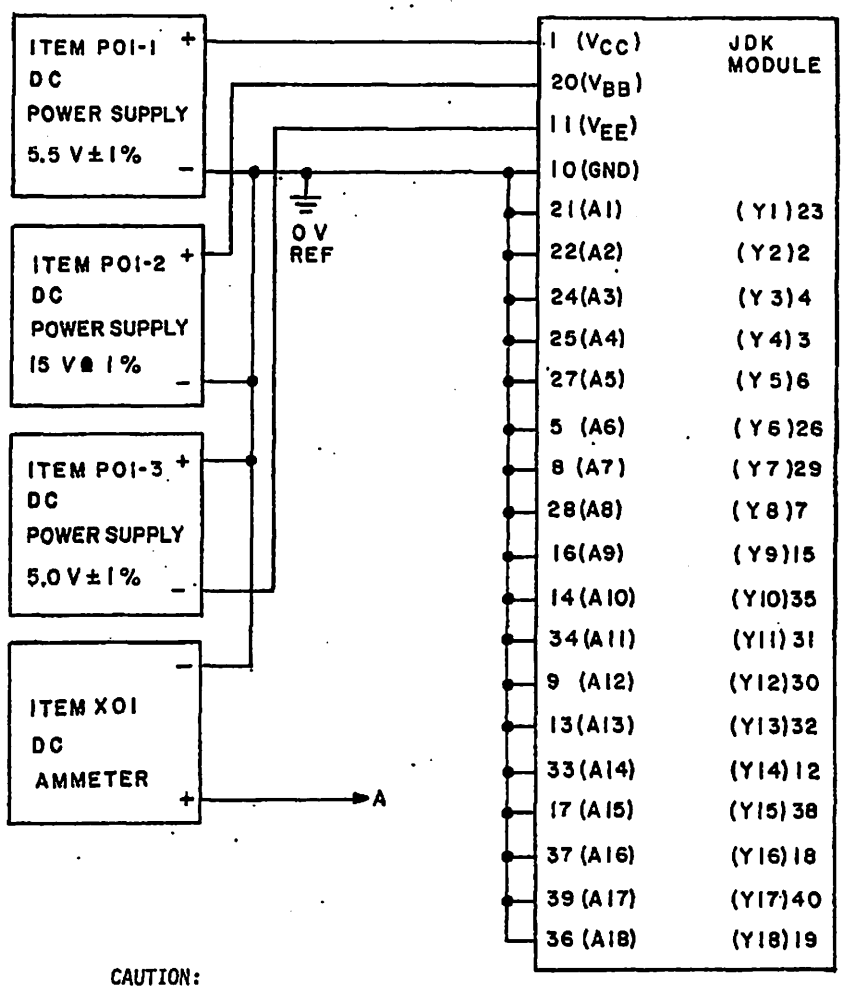
FIGURE 12. Input current (high-level) test circuit for table X.

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TABLE XI. Output short circuit current test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
J1	Y1 Output short circuit current	23	23	55	mA	13
J2	Y2	2				
J3	Y3	4				
J4	Y4	3				
J5	Y5	6				
J6	Y6	26				
J7	Y7	29				
J8	Y8	7				
J9	Y9	15				
J10	Y10	35				
J11	Y11	31				
J12	Y12	30				
J13	Y13	32				
J14	Y14	12				
J15	Y15	38				
J16	Y16	18				
J17	Y17	40				
J18	Y18 Output short circuit current	19	23	55	mA	13

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CAUTION:
Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 13. Output short circuit current test circuit for table XI.

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TABLE XII. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
K1	V_{CC} Power supply current, low level	C	1	A	Open	20	Open	11	30	150	mA	14
K2	V_{CC} Power supply current, high level	C	1	A	F	20	D		5	30		
K3	V_{BB} Power supply current, low level	E	20	1	Open	A	Open		10	58		
K4	V_{BB} Power supply current, high level	E	20		F	A	D	11	15	84		
K5	V_{EE} Power supply current, low level	11	G		Open	20	Open	B	15	84		
K6	V_{EE} Power supply current, high level	11	G	1	F	20	D	B	15	84	mA	14

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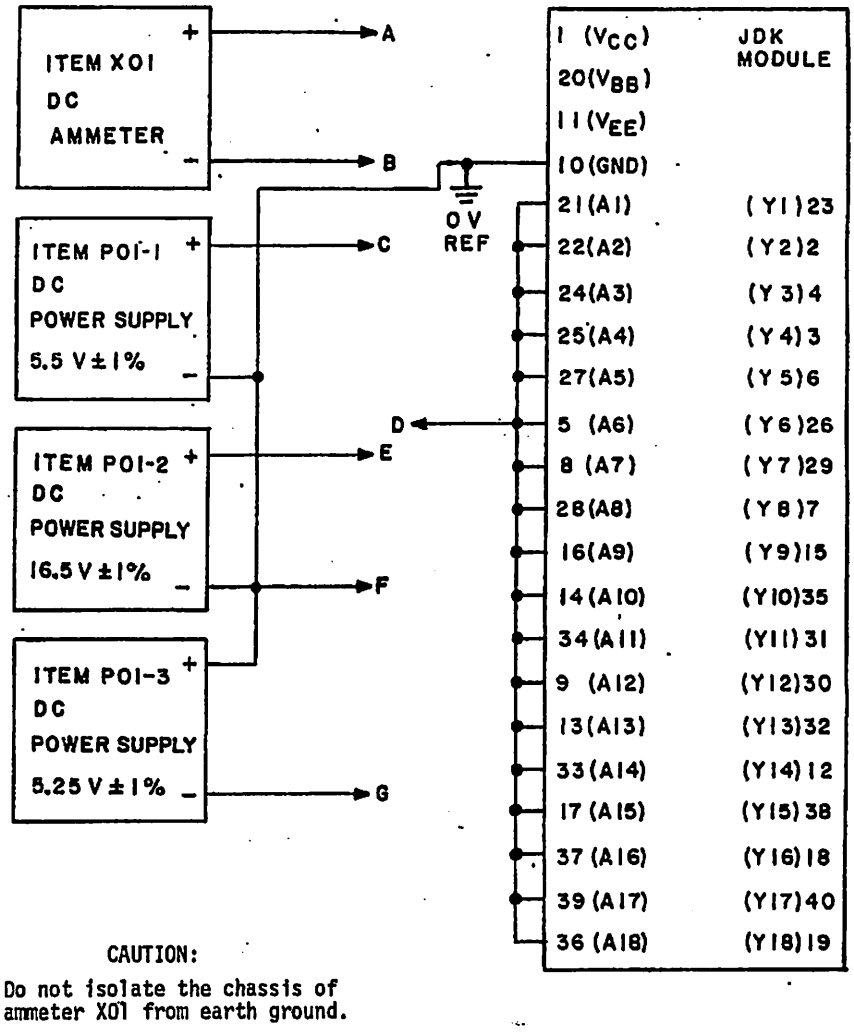


FIGURE 14. Power supply current (low-level and high-level) test circuit for table XII.

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) test.

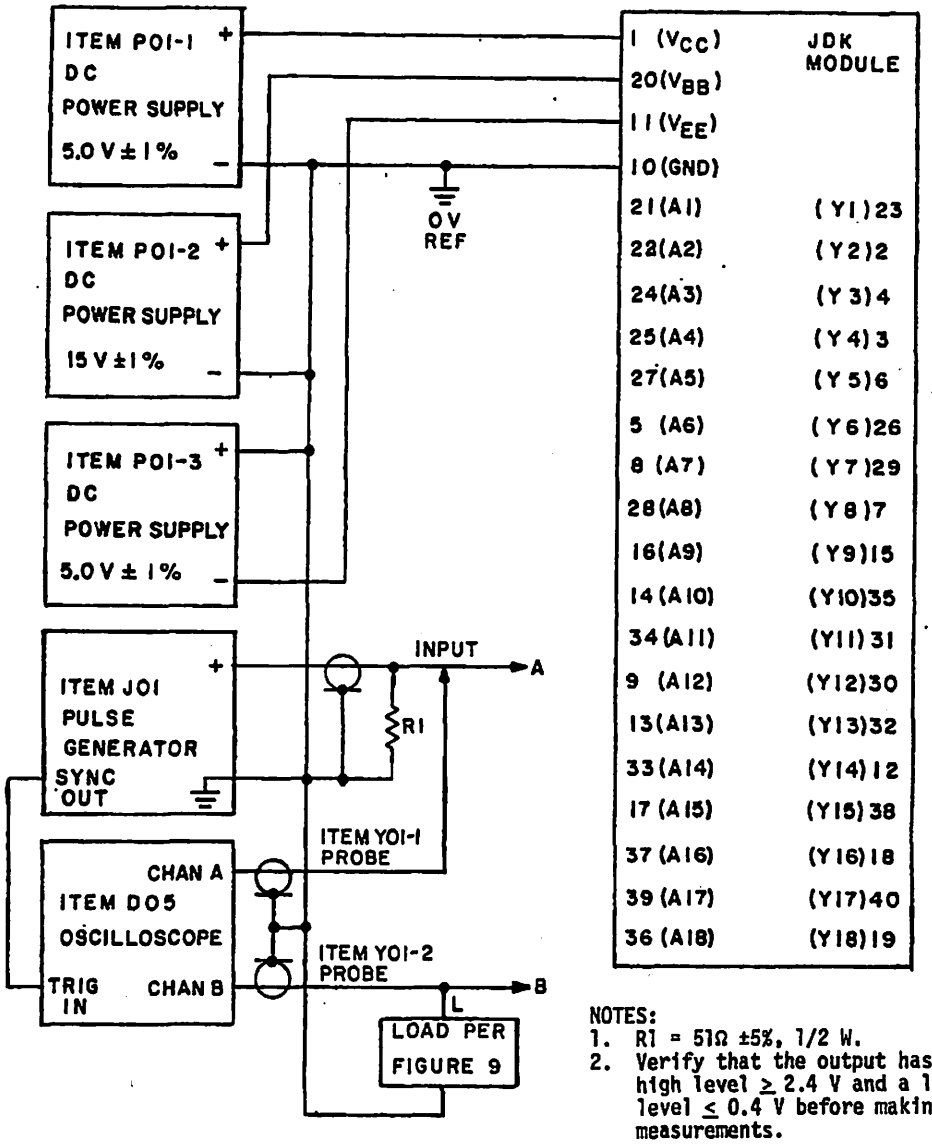
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient				Unit	Test figure
		A to pin	B to pin	t_{TLH}		t_{THL}			
				Min	Max	Min	Max		
L1, M1	Y1 Transition time	21	23	20	72	8	30	ns	15
L2, M2	Y2	22	2						
L3, M3	Y3	24	4						
L4, M4	Y4	25	3						
L5, M5	Y5	27	6						
L6, M6	Y6	5	26						
L7, M7	Y7	8	29						
L8, M8	Y8	28	7						
L9, M9	Y9	16	15						
L10, M10	Y10	14	35						
L11, M11	Y11	34	31						
L12, M12	Y12	9	30						
L13, M13	Y13	13	32						
L14, M14	Y14	33	12						
L15, M15	Y15	17	38						
L16, M16	Y16	37	18						
L17, M17	Y17	39	40						
L18. M18	Y18 Transition time	36	19	20	72	8	30	ns	15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:27Z
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TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient				Unit	Test figure
		A to pin	B to pin	t_{PHL}		t_{PLH}			
				Min	Max	Min	Max		
N1, P1	Y1 Propagation delay time	21	23	10	35	10	30	ns	15
N2, P2	Y2	22	2						
N3, P3	Y3	24	4						
N4, P4	Y4	25	3						
N5, P5	Y5	27	6						
N6, P6	Y6	5	26						
N7, P7	Y7	8	29						
N8, P8	Y8	28	7						
N9, P9	Y9	16	15						
N10, P10	Y10	14	35						
N11, P11	Y11	34	31						
N12, P12	Y12	9	30						
N13, P13	Y13	13	32						
N14, P14	Y14	33	12						
N15, P15	Y15	17	38						
N16, P16	Y16	37	18						
N17, P17	Y17	39	40						
N18, P18	Y18 Propagation delay time	36	19	10	35	10	30	ns	15

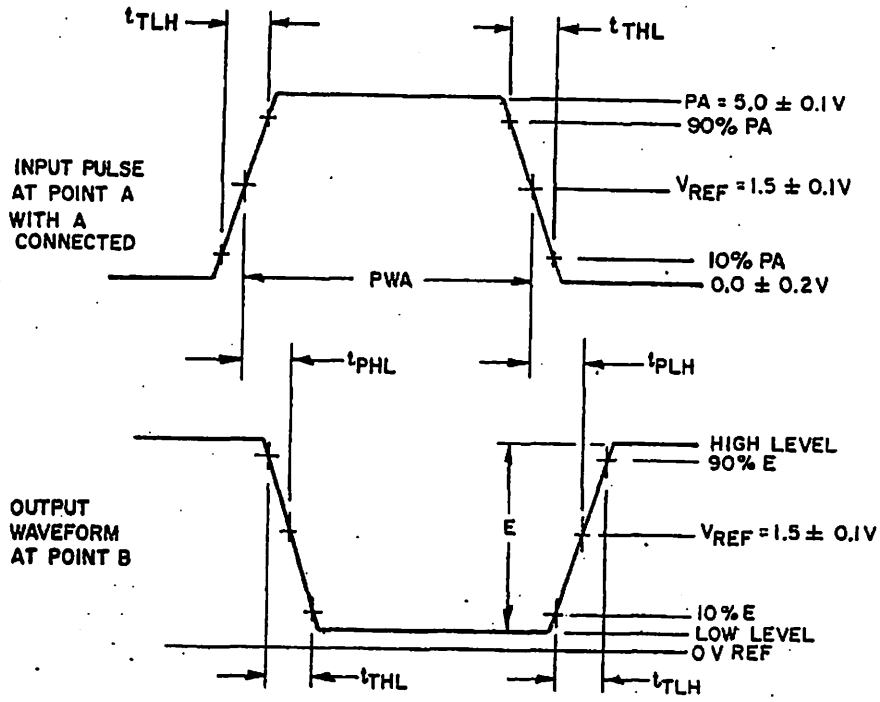
Source: https://assist.dla.mil -- Downloaded: 2026/0-20T15:27Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Verify that the output has a high level ≥ 2.4 V and a low level ≤ 0.4 V before making measurements.

FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit and waveforms for tables XIII and XIV.

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NOTES:
1. Input pulse parameters:
   PWA = 200 ±20 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PRR = 1 MHz ±10%.
2. All voltage levels are referenced to ground.

FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit and waveforms for tables XIII and XIV - Continued.

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TABLE XV. Output crosstalk (low-level) test.

Test cond	Parameter		Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25° ±5°C ambient: Min	Initial limits 25° ±5°C ambient: Max	Unit	Test figure
R1	Y1	Output crosstalk, low level	22,24,25,27,5,8,28,16,14,34, 9,13,33,17,37,39,36	Open	21	23	0	820	mV	16
R2	Y2		21,24,25,27,5,8,28,16,14,34, 9,13,33,17,37,39,36		22	2				
R3	Y3		21,22,25,27,5,8,28,16,14,34, 9,13,33,17,37,39,36		24	4				
R4	Y4		21,22,24,27,5,8,28,16,14,34, 9,13,33,17,37,39,36		25	3				
R5	Y5		21,22,24,25,5,8,28,16,14,34, 9,13,33,17,37,39,36		27	6				
R6	Y6		21,22,24,25,27,8,28,16,14,34, 9,13,33,17,37,39,36		5	26				
R7	Y7		21,22,24,25,27,5,28,16,14, 34,9,13,33,17,37,39,36		8	29				
R8	Y8		21,22,24,25,27,5,8,16,14, 34,9,13,33,17,37,39,36		28	7				
R9	Y9		21,22,24,25,27,5,8,28,14, 34,9,13,33,17,37,39,36		16	15				
R10	Y10		21,22,24,25,27,5,8,28,16, 34,9,13,33,17,37,39,36		14	35				
R11	Y11		21,22,24,25,27,5,8,28,16, 14,9,13,33,17,37,39,36		34	31				
R12	Y12		21,22,24,25,27,5,8,28,16,14, 34,13,33,17,37,39,36		9	30				
R13	Y13		21,22,24,25,27,5,8,28,16,14, 34,9,33,17,37,39,36		13	32				
R14	Y14		21,22,24,25,27,5,8,28,16,14, 34,9,13,17,37,39,36		33	12				
R15	Y15	Output crosstalk, low level	21,22,24,25,27,5,8,28,16,14, 34,9,13,33,37,39,36	Open	17	38	0	820	mV	16

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TABLE XV. Output crosstalk (low-level) test -Continued.

Test cond	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R16	Y16 Output crosstalk, low level	21,22,24,25,27,5,8,28,16,14, 34,9,13,33,17,39,36	Open	37	18	0	820	mV	16
R17	Y17 Output crosstalk, low level	21,22,24,25,27,5,8,28,16,14, 34,9,13,33,17,37,36	Open	39	40	0	820	mV	16
R18	Y18 Output crosstalk, low level	21,22,24,25,27,5,8,28,16,14, 34,9,13,33,17,37,39	Open	36	19	0	820	mV	16

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TABLE XVI. Output crosstalk (high-level) test.

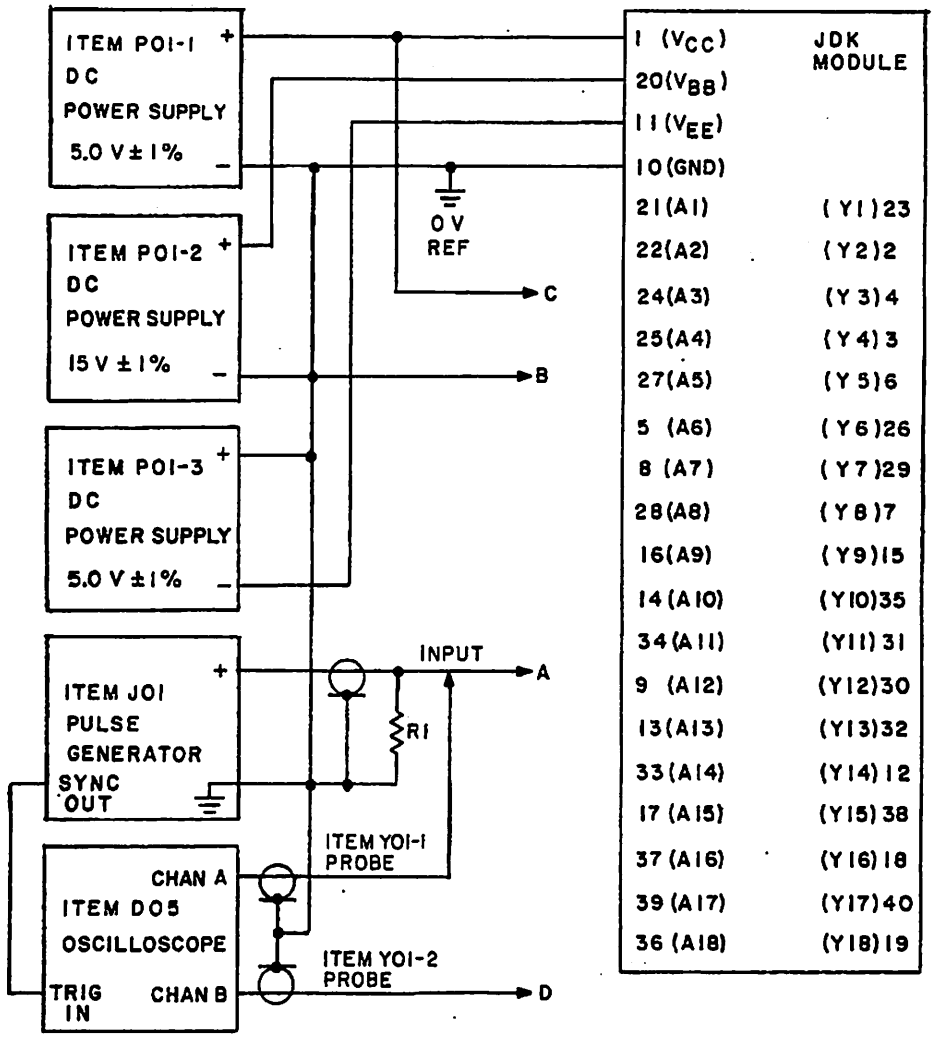
Test cond	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25° ±5°C ambient: Min	Initial limits 25° ±5°C ambient: Max	Unit	Test figure
S1	Y1 Output crosstalk, high level	22,24,25,27,5,8,28,16,14,34, 9,13,33,17,37,39,36	21	Open	23	0	400	mV	16
S2	Y2	21,24,25,27,5,8,28,16,14,34, 9,13,33,17,37,39,36	22		2				
S3	Y3	21,22,25,27,5,8,28,16,14,34, 9,13,33,17,37,39,36	24		4				
S4	Y4	21,22,24,27,5,8,28,16,14,34, 9,13,33,17,37,39,36	25		3				
S5	Y5	21,22,24,25,5,8,28,16,14,34, 9,13,33,17,37,39,36	27		6				
S6	Y6	21,22,24,25,27,8,28,16,14, 34,9,13,33,17,37,39,36	5		26				
S7	Y7	21,22,24,25,27,5,28,16,14, 34,9,13,33,17,37,39,36	8		29				
S8	Y8	21,22,24,25,27,5,8,16,14,34, 9,13,33,17,37,39,36	28		7				
S9	Y9	21,22,24,25,27,5,8,28,14,34, 9,13,33,17,37,39,36	16		15				
S10	Y10	21,22,24,25,27,5,8,28,16,34, 9,13,33,17,37,39,36	14		35				
S11	Y11	21,22,24,25,27,5,8,28,16,14, 9,13,33,17,37,39,36	34		31				
S12	Y12	21,22,24,25,27,5,8,28,16,14, 34,13,33,17,37,39,36	9		30				
S13	Y13	21,22,24,25,27,5,8,28,16,14, 34,9,33,17,37,39,36	13		32				
S14	Y14	21,22,24,25,27,5,8,28,16,14, 34,9,13,17,37,39,36	33		12				
S15	Y15 Output crosstalk, high level	21,22,24,25,27,5,8,28,16,14, 34,9,13,33,37,39,36	17	Open	38	0	400	mV	16

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TABLE XVI. Output crosstalk (high-level) test -Continued.

Test cond	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S16	Y16 Output crosstalk, high level	21,22,24,25,27,5,8,28,16,14, 34,9,13,33,17,39,36	37	Open	18	0	400	mV	16
S17	Y17 Output crosstalk, high level	21,22,24,25,27,5,8,28,16,14, 34,9,13,33,17,37,36	39	Open	40	0	400	mV	16
S18	Y18 Output crosstalk, high level	21,22,24,25,27,5,8,28,16,14, 34,9,13,33,17,37,39	36	Open	19	0	400	mV	16

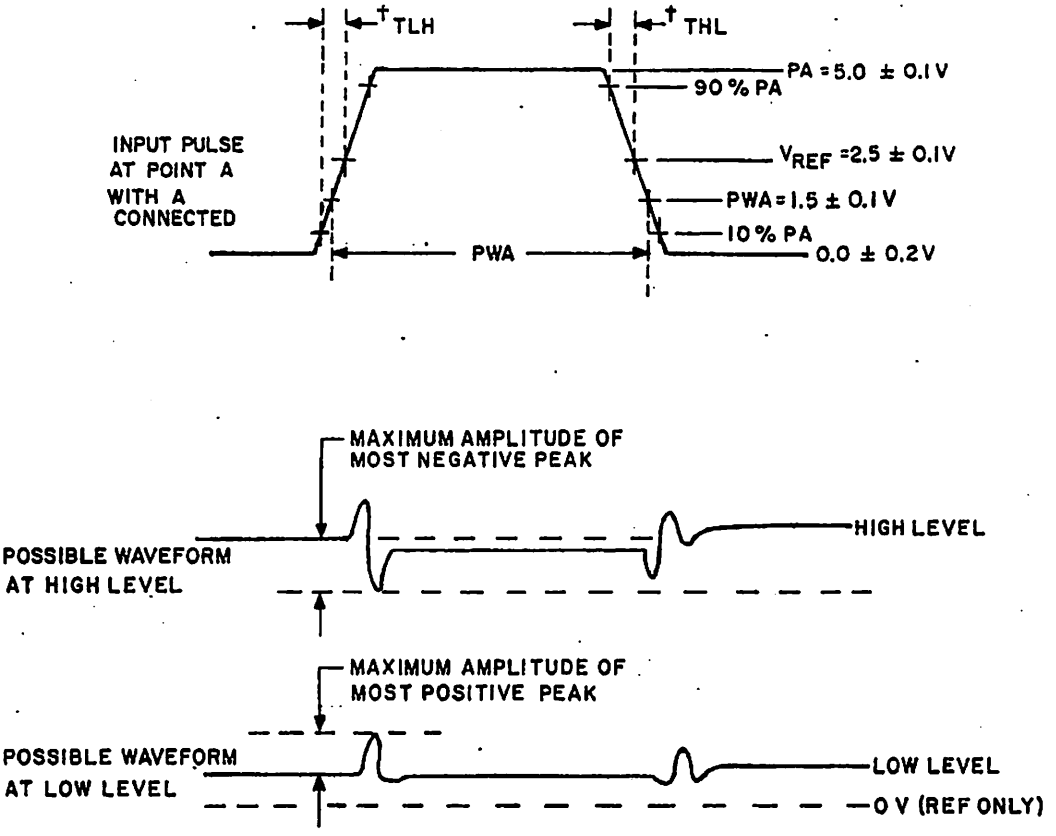
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Load each output with a load as specified on figure 9.

FIGURE 16. Output crosstalk (low-level and high-level) test circuit and waveforms for tables XV and XVI.

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NOTES:
1. Input pulse parameters:
   PWA = 200 ±20 ns.
   $t_{TLH} = t_{THL}$ = 12.5 ±2.5 ns (verified at each input pin).
   PRR = 1 MHz ±10%.
2. Due to increased pulse generator loading, the input dc level may require adjustment.
3. All voltage levels are referenced to ground.

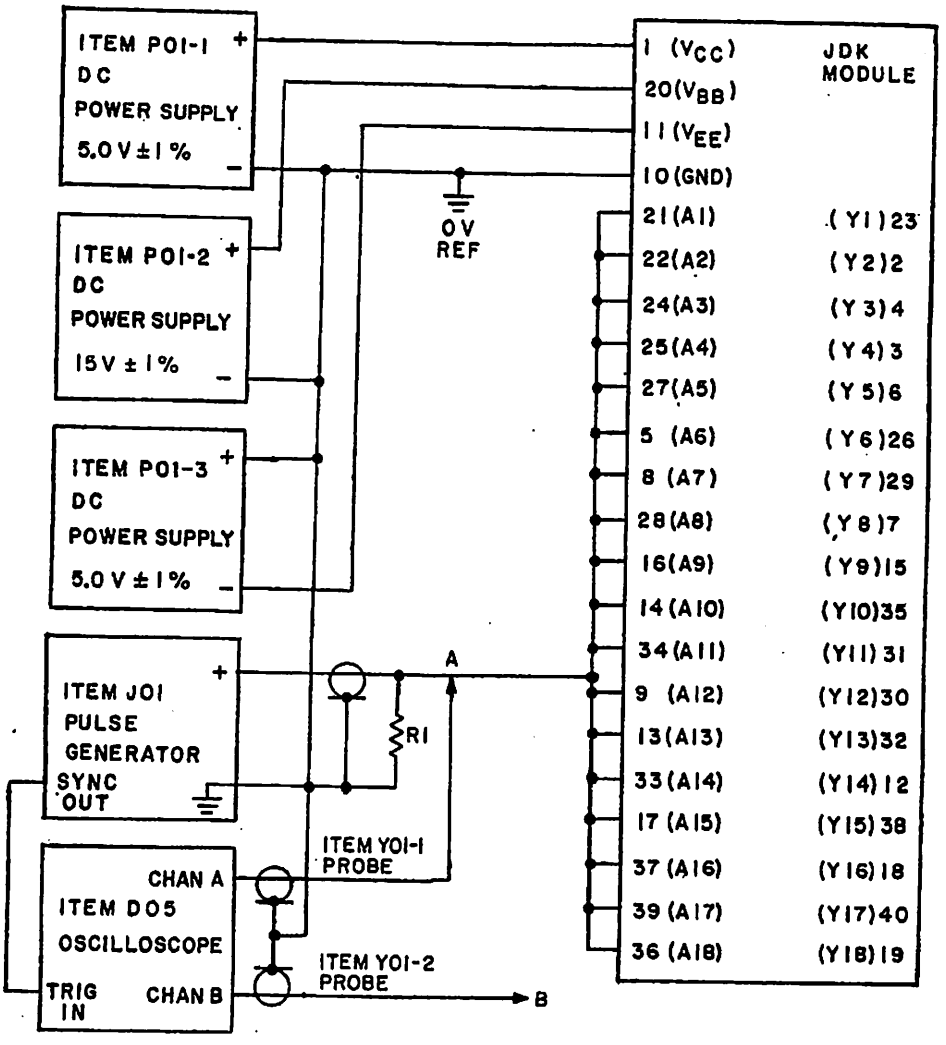
FIGURE 16. Output crosstalk (low-level and high-level) test circuit and waveforms for tables XV and XVI - Continued.

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TABLE XVII. Power supply transient amplitude test.

Test cond.	Parameter	Connect B to pin	Initial limits 25° ±5°C ambient		Unit	Test figure
			Min	Max		
T1	Power supply transient amplitude (V_{CC})	1	0	1.0	V	17
T2	Power supply transient amplitude (V_{BB})	20	0	400	mV	17
T3	Power supply transient amplitude (V_{EE})	11	0	400	mV	17

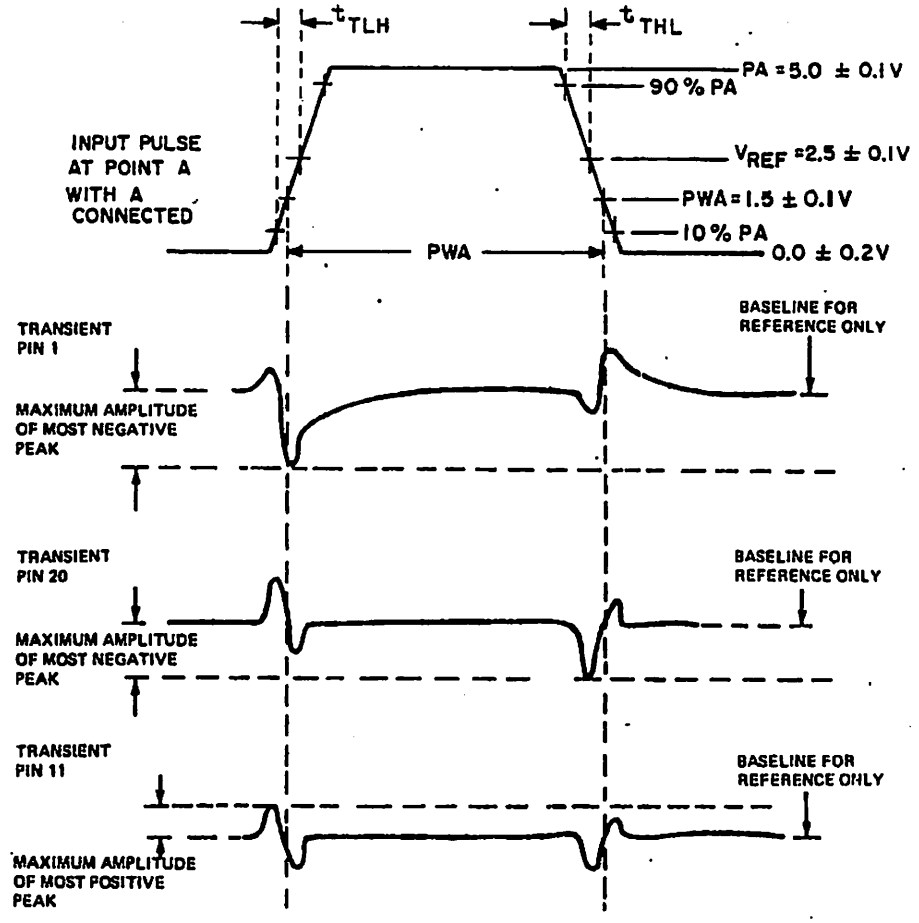
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NOTE: R1 = 51Ω ±5%, 1/2 W.

FIGURE 17. Power supply transient amplitude test circuit and waveforms for table XVII.

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NOTES:
1. Input pulse parameters:
   PWA = 200 ±20 ns.
   $t_{TLH} = t_{THL}$ = 5 ±1 ns (verified at each input pin).
   PRR = 1 MHz ±10%.
2. Due to increased pulse generator loading, the input dc level may require adjustment.

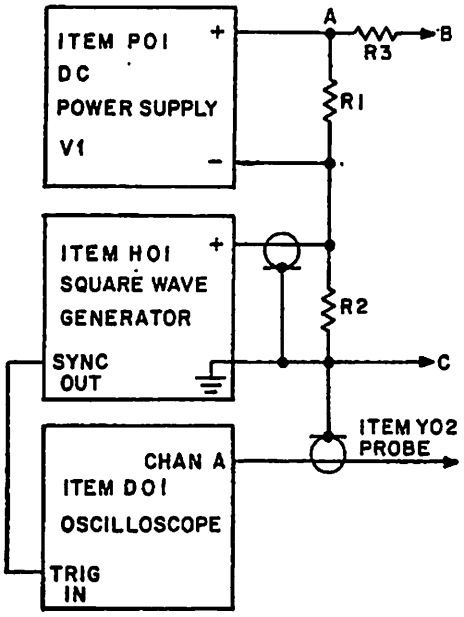
FIGURE 17. Power supply transient amplitude test circuits and waveforms for table XVII - Continued.

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TABLE XVIII. Power supply transient rise time test.

Test cond.	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		B to pin	C to pin	Min	Max		
U1	Power supply transient rise time (V_{CC})	1	10	1.0	2.0	ms	18
U2	Power supply transient rise time (V_{BB})	20	10	1.0	2.0	ms	18
U3	Power supply transient rise time (V_{EE})	← See figure 19 →				ms	19

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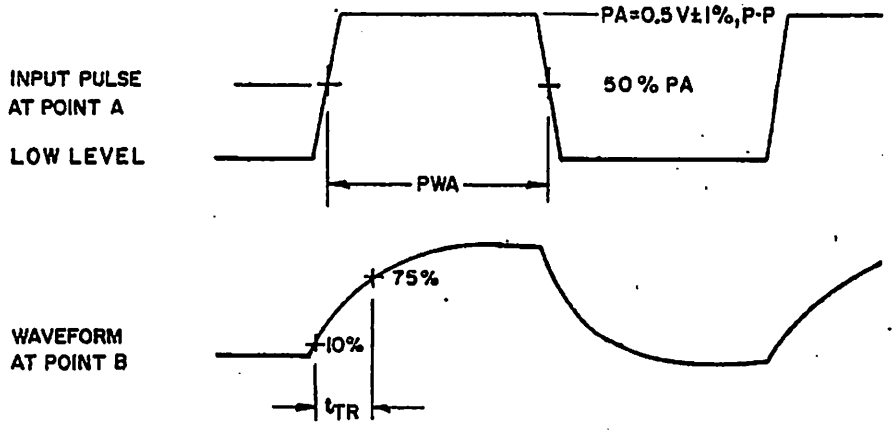


	JDK MODULE
1 (V_{CC})	
20(V_{BB})	
11(V_{EE})	
10(GND)	
21(A1)	(Y1)23
22(A2)	(Y2)2
24(A3)	(Y3)4
25(A4)	(Y4)3
27(A5)	(Y5)6
5 (A6)	(Y6)26
8 (A7)	(Y7)29
28(A8)	(Y8)7
16(A9)	(Y9)15
14(A10)	(Y10)35
34(A11)	(Y11)31
9 (A12)	(Y12)30
13(A13)	(Y13)32
33(A14)	(Y14)12
17 (A15)	(Y15)38
37 (A16)	(Y16)18
39 (A17)	(Y17)40
36 (A18)	(Y18)19

NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 511Ω ±1%, 1/4 W.

FIGURE 18. Power supply transient rise time test circuit and waveforms for table XVIII.

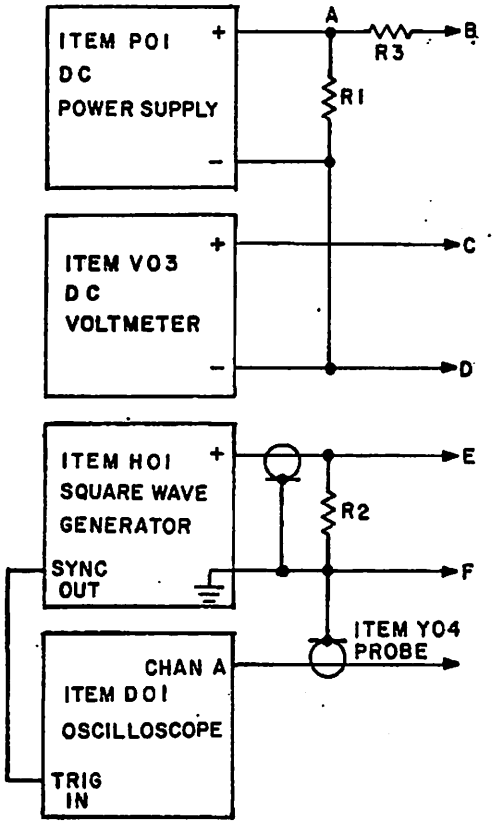
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NOTES:
1. Input pulse parameters:
   PWA = 10 ms ±10%.
   PRR = 50 Hz (reference).
   Rise and fall time of input < 100 ns (reference).
2. With point "B" open, adjust input at point "A" as shown.
3. With point "B" open, adjust V1 such that the low level of the input is at 0 ±0.01 V.
4. Connect points "B" and "C" as defined in table XVIII. After making connections for an electrical requirement, recheck low level and input pulse parameters at point "A".
5. Measure (t_{TR}) at point "B".

FIGURE 18. Power supply transient rise time test circuit and waveforms for table XVIII - Continued.

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	JDK MODULE
1 (V_{CC})	
20(V_{BB})	
11(V_{EE})	
10(GND)	
21(A1)	(Y1)23
22(A2)	(Y2)2
24(A3)	(Y3)4
25(A4)	(Y4)3
27(A5)	(Y5)6
5 (A6)	(Y6)26
8 (A7)	(Y7)29
28(A8)	(Y8)7
16(A9)	(Y9)15
14(A10)	(Y10)35
34(A11)	(Y11)31
9 (A12)	(Y12)30
13(A13)	(Y13)32
33(A14)	(Y14)12
17 (A15)	(Y15)38
37 (A16)	(Y16)18
39 (A17)	(Y17)40
36 (A18)	(Y18)19

NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 511Ω ±1%, 1/4 W.
3. Adjust item P01 for 0.25 V ±1% at point "A" with points "B" and "C" open.
4. Connect points "B" and "C" to module pin 10. Connect point "D" to module pin 11.
5. The value measured on item V03 shall be recorded as V1 in volts under note 7 below.
6. Disconnect points "B", "C", and "D".
7. Calculate and record in the appropriate columns of table XVIII, the following limits:

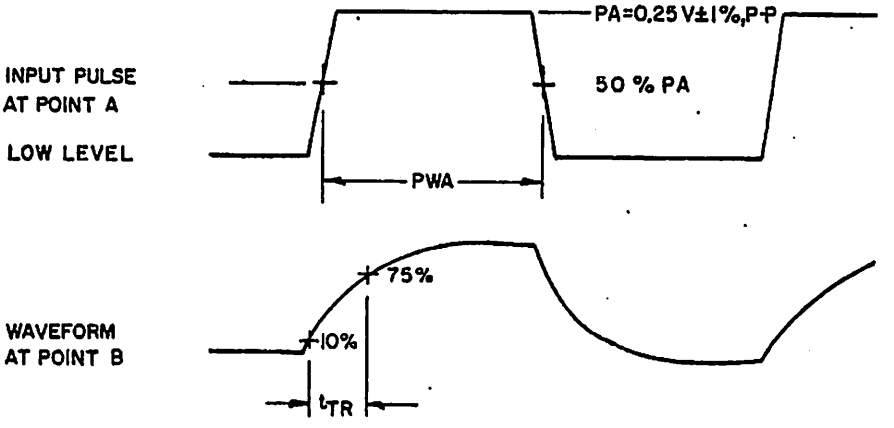
   a. t_{TR} minimum = 4 V1 = 4 ( ) = _____ ms.

   b. t_{TR} maximum = 8 V1 = 8 ( ) = _____ ms.

8. Once a design has been approved, a minimum and maximum limit calculation can be made. This limit can then be used whenever testing that particular design.

FIGURE 19. Power supply transient rise time test circuit and waveforms for table XVIII.

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NOTES:
1. Connect point "D" to point "E". Connect point "F" to module pin 11.
2. With point "B" open, adjust the input at point "A" as follows:
   PWA = 10 ms ±10%.
   PRR = 50 Hz (reference).
   Rise time and fall time < 100 ns (reference).
3. With point "B" open, adjust item P01 such that the low level of the input at point "A" is 0 ±0.01 V.
4. Connect point "B" to pin 10. After making connection, recheck the low level and input pulse parameters at point "A".
5. Measure t_{TR} at point "B".

FIGURE 19. Power supply transient rise time test circuit and waveforms for table XVIII - Continued.

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